

GOLDEN TEE 2005

Complete Restoration Guide for Public Placements
Cabinet • Controls • Electronics • Monitor • Reliability

Prepared for **Coin Op 4 Charity** / Arcade Angels

Practical notes so restored machines stay reliable in youth centers and community spaces

1. Overview & Goals for Public Machines

Golden Tee 2005 (Incredible Technologies) is one of the strongest draws for casual play in youth centers and community venues. A complete restoration for public use prioritizes three things above pure originality: **reliability**, **safety**, and **player experience**. This guide walks through the whole machine so a restored unit can be placed with confidence and stay in service with minimal call-backs.

2. Safety First

- Always unplug the cabinet before opening it.
- CRT monitors hold high voltage after power-off. Discharge the anode correctly before working on the chassis or neckboard.
- If you are not experienced with CRTs, strongly consider an LCD conversion for any machine going into a youth or public space.
- Keep the work area clean and organized — small parts and screws are easy to lose and can create hazards if left inside the cabinet.

3. Cabinet Exterior & Cosmetics

First impressions matter for public placements. A clean, intact cabinet signals that the machine is cared for.

- Wash the exterior with mild cleaner. Remove sticky residue, marker, and years of hand prints.
- Inspect side art, front decals, and control-panel overlay. Replace or touch up heavily damaged graphics if they make the machine look neglected.
- Check the coin door, locks, and hinges. Lubricate lightly and confirm keys work or replace locks if needed for the placement site.
- Marquee: clean the plastic, replace fluorescent tubes or starters if dim or flickering. A bright marquee makes the machine far more inviting.
- Level the cabinet and confirm it sits solidly — no rocking.

4. Control Panel, Trackball & Buttons

Trackball (highest priority)

A sticky or dirty trackball is the number-one player complaint. Remove the trackball, clean the ball, rollers, and optical sensors thoroughly. Inspect the bearings/rollers for flat spots or wear. Re-lubricate sparingly with a plastic-safe lubricant. Test for smooth, accurate movement in all directions before buttoning the panel back up.

Buttons & switches

Clean or replace sticky or unresponsive buttons. Confirm every button (including Start and any extra function buttons) registers cleanly in test mode. Replace worn or cracked button tops so the panel looks and feels intentional.

Control panel overlay & surface

If the overlay is heavily scratched, torn, or peeling, a replacement improves both appearance and hygiene for public use. Secure any loose edges so they cannot lift further.

5. Power Supply & Voltage

Golden Tee boards (especially green boards) are sensitive to low or unstable +5 V. Measure voltage at the board under load. Aim for roughly 5.1–5.2 V. Adjust the supply if it sits low. Confirm all required power connectors are fully seated —

these boards often use multiple connectors. A weak supply causes freezes, boot loops, and intermittent color or video problems that look like other failures.

6. Game Board (Green / Red) Notes

- Reseat the video card. Clean the contacts; oxidation is common and can cause boot or video failures.
- Inspect for corrosion (more common on green boards). Clean carefully if present.
- Confirm status LEDs behave as expected during boot (refer to community documentation for exact LED meanings on your board revision).
- If the board has a known history of freezes or random reboots, check capacitors and voltage rails. Many later operators convert storage and clean power issues before chasing deeper board faults.

7. Hard Drive / Storage Conversion

Original spinning hard drives fail regularly from heat and age. Converting to CompactFlash or an SSD with a proper IDE adapter is one of the highest-value reliability upgrades for public machines. Benefits: faster boots, less heat, far fewer mechanical failures. Keep a verified backup image of the original drive contents before and after conversion. Label the new media clearly with the software version / courses present.

8. Monitor — CRT Service or LCD Conversion

Green hue / color problems (CRT path)

Common causes: cold solder or lifted traces on the neckboard color-drive transistors, weak green drive transistor, loose RGB connectors, drifted color pots, or video harness issues. Start with reseating cables and the neckboard, then reflow the three color transistors and RGB header on the neckboard. Many machines regain correct color balance from a careful reflow alone. Adjust drive/cutoff pots afterward for neutral whites and accurate greens.

LCD conversion (recommended for youth / high-use sites)

Advantages for public placements: lower heat, no high-voltage hazard, consistent picture, easier long-term support. Golden Tee 2005 does not output standard VGA cleanly — a converter board (GBS-8200 style or similar) is usually required. DIP switches on the game board select standard vs medium resolution; test both. Commercial kits with brackets and bezels exist (Happ/Suzo-Happ, Unico, Makvision, etc.). Some cabinet woodwork or shimming is often needed. Preserve the original CRT and chassis labeled and stored if possible — useful as a donor or for future restoration.

9. Wiring, Coin Door & Miscellaneous

- Inspect major harnesses for cracked insulation or strain at connectors. Repair or reinforce as needed.
- Coin door: clean switches, confirm coin lockout and coin counter (if used) function. For free-play youth placements, set the machine to free play or configure credits appropriately and document the setting.
- Speakers: clean, check for torn cones, and confirm both channels work at reasonable volume.
- Cooling: clear dust from any fans and vents. Heat shortens the life of boards and drives.

10. Final Testing Checklist Before Placement

- Full cold boot to attract mode with correct colors and no dominant green cast
- Trackball smooth and accurate in all directions
- Every button and coin switch registers cleanly
- Sound present and balanced from both speakers
- Complete at least one full 9-hole (preferably 18-hole) game with no freezes or reboots
- Monitor geometry, focus, and brightness acceptable from normal player stance
- Cabinet exterior clean, level, and free of sharp edges or loose parts
- Marquee lit and readable
- Free-play or pricing configured correctly for the placement site
- Keys, manuals, and any spare parts bagged and labeled for the host organization

11. Notes for Host Sites (Youth Centers & Community Spaces)

- Provide a short one-page instruction sheet: how to power on/off, free-play status, who to call for problems.
- Recommend a simple weekly wipe-down of the trackball and control panel.

- Ask the host to report any new stickiness, freezes, or picture changes early so small issues stay small.
- Document the machine's software version / courses and any non-original parts (LCD, CF card, etc.) so future volunteers know what they are looking at.

This guide is written for educational and nonprofit restoration use by Coin Op 4 Charity and partner organizations. The goal is machines that stay reliable, safe, and fun for the people who actually play them.

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